

Total Surface Area of a Cuboid: $1b \times b$

 $2 \times (lb + bh + lh)$

double ts-a-cuboid (double l, double b, double h) {
 (operations:)
 return ts-a;
}

*
 ↑
 ↓

Create a C function to perform the same.

off = 321

int reverseInteger (int ¹²³n) {

}

→ 321;

ans = 321

123 % 10 = 3
 123 / 10 = 12
 12 % 10 = 2
 12 / 10 = 1
 1 % 10 = 1
 1 / 10 = 0

stop

IT $\rightarrow$ 12-16 $D \rightarrow$ 1/1m $\leftarrow$ TL $\rightarrow$ 3L
 2015 $\rightarrow$ school $\leftarrow$ WHI $\leftarrow$ M $\leftarrow$ 24 $\leftarrow$ 70-80k
Scratch $\leftarrow$ 3L/Std $\leftarrow$ Bxus $\leftarrow$ POM $\leftarrow$ 10LPA
 BDC

int rev(int n)

```
{  
    ans = 0;  
    while(n > 0){  
        int dig = n % 10;  
        ans = ans * 10 + dig;  
        n /= 10;  
    }  
    return ans;  
}
```

template

ans = 0

ans = ans * 10 + digit

= 0 + 3

ans * 10 + d = 30 + 2 = 32

ans * 10 + d = 320 + 1 = 321

dig = 123 % 10 = 3

123 / 10 = 12

12 % 10 = 2

12 / 10 = 1

1 % 10 = 1

1 / 10 = 0

2 * 10 + 1 = 21

32 * 10 + 1 = 321

* No of 1's
* No of set bits

HW D-B ()

I, I'
B, B' [Bitwise operator]

int hammingWeight (int n) {
int count = 0;
Logic
return count; }

11
3
3

$n = 11 \rightarrow 1011 \rightarrow$
 $3 \rightarrow e/o \% \boxed{1}$
 $\downarrow$
 $\boxed{0011 \& 1}$
 0011
 0001
 $\hline 0001 = 1$
 $while (n > 0) \{$
 $\quad if (n \& 1) \{$
 $\quad \quad cut ++;$
 $\quad n = n >> 1$
 $\quad \}$
 $\}$
 $return cut;$

11 $\rightarrow$ 1011
(n & 1) while (n > 0) {
 1011
 0001
 0001 $\rightarrow$ T
 101 & 1
 001
 001
 10
 01
 00
 if (n & 1) {
 cnt++;
 n = n >> 1;
 }
}
return cnt;

Pointers $\rightarrow$

int a = 10;

variable [memory]
[address]

value 10

printf("%d", &a);

address operator

ptr

reference

int *ptr = &a;

2000

3000

address into another variable *double pointer

Handwritten diagram showing the transformation of a program fragment. The initial code is:

```

a = 2000;
ptr = 3000;
dptr = 3000;

```

The transformation steps are indicated by red arrows and labels:

- Step 1: a is replaced by $2a$ (labeled $2a$).
- Step 2: ptr is replaced by 3000 (labeled 3000).
- Step 3: $dptr$ is replaced by ptr (labeled $dptr = ptr = 3000$).

The final code is:

```

2a;
ptr = 3000;
ptr = 3000;

```